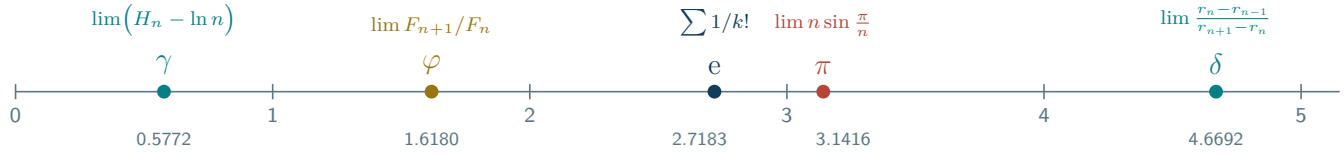
FIVE GREAT CONSTANTS FROM 0 TO 5 ON ONE PAGE

γ, φ, e, π, δ, one in each unit interval

Each is the limit of a simple repeated process, from a root of a quadratic to numbers not yet proved irrational.

THE RULE

Every one of them is a limit



BETWEEN 0 AND 1

Euler–Mascheroni $\gamma = 0.577\,215\,66\ldots$

The harmonic sum outruns $\ln n$ by a gap that settles, and splitting it term by term gives a convergent series and an integral:

$$\gamma = \sum_{k\geq 1}\left(\frac{1}{k} - \ln\left(1+\frac{1}{k}\right)\right) = -\int_0^\infty \mathrm{e}^{-x}\ln x\,\mathrm{d}x.$$

It has no known trigonometric form, and it also appears as $-\Gamma'(1)$.

BETWEEN 1 AND 2

Golden ratio $\varphi = 1.618\,033\,98\ldots$

The positive root of $x^2 = x + 1$, so $x = 1 + \frac{1}{x}$ repeated gives the continued fraction $[1;1,1,1,\ldots]$ and the Fibonacci ratios $\frac{2}{1}, \frac{3}{2}, \frac{5}{3}, \frac{8}{5}, \ldots$. In trigonometric form it is the diagonal of a regular pentagon of side 1,

$$\varphi = \frac{1+\sqrt{5}}{2} = 2\cos\frac{\pi}{5} = 2\cos 36^\circ.$$

BETWEEN 2 AND 3

Euler's number $\mathrm{e} = 2.718\,281\,83\ldots$

Compound growth and factorials give the same limit,

$$\mathrm{e} = \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n = \sum_{k\geq 0}\frac{1}{k!} = 1 + 1 + \tfrac{1}{2} + \tfrac{1}{6} + \ldots,$$

where eight terms already give 2.71825. Its trigonometric face is $\mathrm{e}^{\mathrm{i}\theta} = \cos\theta + \mathrm{i}\sin\theta$, which ties it to π through $\mathrm{e}^{\mathrm{i}\pi} = -1$.

BETWEEN 3 AND 4

Archimedes' constant $\pi = 3.141\,592\,65\ldots$

Inscribed polygons give $n\sin\frac{\pi}{n}$, or $n\sin\frac{180^\circ}{n}$, which is 3 for a hexagon, 3.1058 for a 12-gon and 3.1410 for Archimedes' 96-gon. Arctangent series give it exactly,

$$\frac{\pi}{4} = \arctan 1 = 1 - \tfrac{1}{3} + \tfrac{1}{5} - \cdots = 4\arctan\tfrac{1}{5} - \arctan\tfrac{1}{239},$$

the last being Machin's formula of 1706.

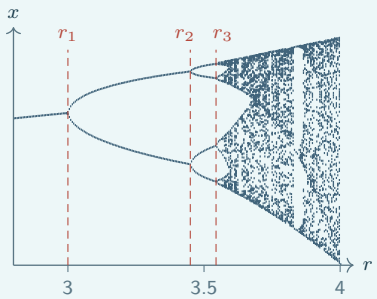
BETWEEN 4 AND 5

Feigenbaum's $\delta = 4.669\,201\,61\ldots$

Iterate the logistic map $x \mapsto rx(1-x)$. Its long-run behaviour settles on one value until $r_1 = 3$, on a cycle of two until $r_2 = 1+\sqrt{6} \approx 3.4495$, of four until $r_3 \approx 3.5441$, of eight until $r_4 \approx 3.5644$, and the gaps shrink by a ratio that converges:

$$\frac{r_2 - r_1}{r_3 - r_2} = 4.751, \quad \frac{r_3 - r_2}{r_4 - r_3} = 4.656, \quad \ldots \quad \to \quad \delta = 4.6692\ldots$$

Feigenbaum found in 1975 that every smooth one-humped map gives the same δ, a universal constant of the route to chaos with no known series or trigonometric form.



Long-run values of the logistic map: each fork arrives about 4.67 times sooner.

From algebraic to unknown

φ	irrational but algebraic, a root of $x^2 - x - 1$, so not transcendental
e and π	transcendental, proved by Hermite in 1873 and Lindemann in 1882
γ and δ	believed transcendental, though neither has yet been proved even irrational

Other constants share these intervals, such as $\sqrt{2} \approx 1.414$ and Apéry's $\zeta(3) \approx 1.202$, proved irrational in 1978 but not yet transcendental, and even $\mathrm{e} + \pi$ is not known to be irrational.

Conventions. A number is algebraic if it solves a polynomial equation with integer coefficients and transcendental otherwise; H_n is the nth harmonic number, F_n the nth Fibonacci number, and r_n the parameter where the logistic cycle first reaches length 2^n.